Dino v0.01 — Build Instructions and Bill of Materials

Objective

Construct a manual, CPU-free logic sequencer capable of stepping through memory and outputting 8-bit values using LEDs. This stage establishes the address counter, SRAM interface, manual program loading, and basic execution loop.

Bill of Materials (Dino v0.01)

Qty	Part	Description
2	SN74LS590N	8-bit binary counters for 16-bit address
1	MB81256-10	32KB SRAM for program/data memory
1	555 Timer	Clock generation (RUN mode)
1	SPDT Switch	LOAD/RUN mode selector
2	Pushbuttons	Manual WRITE and CLOCK control
1	8-bit DIP Switch	Byte input during LOAD mode
8	LEDs	Output display for memory byte
8	10kΩ Resistors	Pull-downs for DIP switch
8	220–470Ω Resistors	LED current limiting
1	Breadboard	Prototyping platform
-	Jumper Wires	Wiring interconnects
1	5V Power Supply	Logic and memory voltage source

Build Steps

Step 1: Program Counter

1. Chain 2× SN74LS590N counters for 16-bit address generation
2. Connect 555 timer output (or manual step button) to CP0 of lower counter
3. Connect RC of lower counter to CP0 of upper counter
4. Output Q0–Q15 → connect to SRAM A0–A15

Step 2: SRAM Wiring

1. Tie SRAM /OE LOW to always enable output
2. Connect /WE to a pushbutton (active LOW) for manual write

3. Route D0–D7 to:
    - DIP switch during LOAD mode
    - LED bank during RUN mode via mode switch or jumper swaps

Step 3: Manual Programming (LOAD Mode)

1. Set mode switch to LOAD
2. Use DIP switch to set 8-bit data value
3. Press WRITE to store value into SRAM
4. Press CLOCK to increment address counter
5. Repeat steps 2–4 for each byte of program

Step 4: Program Execution (RUN Mode)

1. Set mode switch to RUN
2. Start 555 timer (or step manually)
3. Observe LEDs display stored data sequentially

Completion Criteria

- SRAM can be loaded manually with 8-bit values
- Counter steps through addresses properly
- LEDs display expected data from memory
- Prepares system for Phase 2 (register and ALU integration)